Karina Bryant

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Summary

Theoretical physics data analyst, experienced in multimessenger astrophysics signal processing, data analysis, and computational physics and programming. Looking to apply my degree in physics and experience with data science to future career paths in research and industry.

Education

Rochester Institute of Technology — **Aug. 2021 – May 2025**

Bachelor of Science in Pyhsics — *Rochester, NY*

- GPA – 3.1

Relevant Coursework

- Computational Physics and Programming
- Principles of Computing
- Data Science
- Technical Writing
- Cyber Security Policy and Law

Research Projects

Senior Capstone | *Python, Computational Physics, Data Analysis* — **Aug 2024 – May 2025**

- **Emission Line Signatures of Recoiling Supermassive Black Holes.**
- Collaborated with graduate students in RIT's School of Physics and Astronomy to create the first program that produces emission line spectra of quasars from theoretical N-body simulations of recoiling supermassive black holes.
- Those spectra can then be applied and compared to data gathered by the Sloan Digital Sky Survey, one of the most successful surveys in the history of astronomy.
- This code will help further our general understanding of supermassive black hole evolution and expand the tools available to future researchers.

Emerson Fellowship | *Python, Computational Physics* — **May 2024 – August 2024**

- Created and developed the first N-body code that models the effects of supermassive black hole recoils on its surrounding broadline region.
- This versatile python code can simulate supermassive black hole recoils with any given initial conditions over any time-frame.
- Further application of this research will allow for further simulation of emission line signatures and will be used by other researchers and scientists to further study this specific area of astrophysics.

Inclusive Excellence Fellowship | *Python, Machine Learning* — **May 2022 – August 2022**

- Research Project on Gravitational Waves
- Simulated data gathered from radio telescopes using auto-regressive models to study the effects of the interstellar medium on gravitational waves.
- That data is used to model how gravitational waves influence the real data from pulsar timing arrays.

Technical Skills

Languages: Python (NumPy, SciPy, Sklearn, Pandas, Matplotlib, REBOUND), SQL
Developer Tools: VS Code, AstroImageJ, LaTeX
Training and Soft Skills: Data Visualization, Data Analysis, Project Collaboration, Technical Writing, Machine Learning

Community Involvement / Awards

RIT College of Science Undergraduate Research Scholar — **April 2025**

Award

- Award honoring a select group of graduating students who have participated in mentored research projects. These scholars are selected based on the unique demands and competencies required in their respective fields.

FIRST Robotics — **2012 – Present**

Non-profit focused on engaging students in full life cycle engineering through competition.

- Participated from elementary school through high school - 4 years of FIRST Lego League (FLL), 2 years of FIRST Tech Challenge (FTC), and 4 years of FIRST Robotics Competition (FRC).
- Acted as a primary representative for the team as a member of the competition drive team during FTC and FRC and became the strategy team leader during FRC.
- Continued volunteering after graduating, acting as a team mentor and a judge at competitions.